

CLTL at ArAIEval Shared Task: Multimodal Propagandistic Memes Classification Using Transformer Models

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Abstract

We present the CLTL system designed for the ArAIEval Shared Task 2024 on multimodal propagandistic memes classification in Arabic. The challenge was divided into three subtasks: identifying propagandistic content from textual modality of memes (subtask 2A), from visual modality of memes (subtask 2B), and in a multimodal scenario when both modalities are combined (subtask 2C). We explored various unimodal transformer models for Arabic language processing (subtask 2A), visual models for image processing (subtask 2B), and concatenated text and image embeddings using the Multilayer Perceptron fusion module for multimodal propagandistic memes classification (subtask 2C). Our system achieved 77.96% for subtask 2A, 71.04% for subtask 2B, and 79.80% for subtask 2C, ranking 2nd, 1st, and 3rd on the leaderboard.

1 Introduction

In the digital age, memes have gained immense popularity across various age groups through social media platforms such as Facebook and Twitter. In recent years, however, they have been increasingly employed for propagandistic purposes (G. De Leon and Ballesteros-Lintao, 2021; Kingdon, 2021). These memes often deploy a range of propagandistic techniques, including logical fallacies, causal oversimplification, and exaggeration, paired with images with strong positive/negative emotional implications to subtly sway audience opinions and propagate misinformation (Nieuburt, 2021).

Given the multimodal nature and powerful impact of these memes, it is crucial to develop classification systems capable of detecting propagandistic memes effectively. Previous studies in propaganda detection primarily focused on textual content (Barrón-Cedeño et al., 2019; Alam et al., 2022; Hasanain et al., 2023, 2024a) or fine-grained persuasion techniques analysis (Dimitrov et al., 2024).

To the best of our knowledge, the ArAIEval 2024 Shared Task (Hasanain et al., 2024b) is the first shared task on propaganda detection in multimodal scenarios, specifically within Arabic memes. The task is divided into three subtasks: subtasks 2A and 2B focus on classifying the textual and visual modalities of a given meme separately to detect whether it is propagandistic or not, whereas subtask 2C integrates both modalities for multimodal classification to detect whether a meme is propagandistic.

We carried out various experiments using state-of-the-art Arabic language processing models and vision models, and employed the Multilayer Perceptron (MLP) fusion module (Shi et al., 2021) with a prediction layer on top for multimodal classification. Without the need for text preprocessing and feature engineering, our system secured 2nd place in subtask 2A, 1st place in subtask 2B, and 3rd place in subtask 2C.

2 Data

The dataset used in the ArAIEval 2024 Shared Task comprises a collection of Arabic memes sourced from various social media platforms: Facebook, Twitter, Instagram, and Pinterest (Alam et al., 2024). Each meme is labeled as either ‘propagandistic’ or ‘non-propagandistic’ and includes both the original image file and its corresponding extracted textual content. The single dataset was used for the three subtasks covered in the competition.

The statistics of the dataset, in terms of the number of memes per class as well as the class distribution, are provided in Table 1. It can be observed that the dataset is imbalanced in terms of the represented classes, with the propagandistic content constituting the minority class with 28.14%, 28.21%, and 28.17% of the training, development, and test data, respectively.

Table 1: Dataset statistics in terms of the number of memes per class (# Num) and class distribution (%).

Label	Train		Dev		Test	
	# Num	%	# Num	%	# Num	%
propaganda	603	28.14	88	28.21	171	28.17
not_propaganda	1,540	71.86	224	71.79	436	71.83
Total	2,143	100	312	100	607	100

3 Methodology

We conducted comprehensive experiments to evaluate the effectiveness of several state-of-the-art transformer models for Arabic language and image processing to address subtasks 2A and 2B, respectively. For subtask 2C, we used the Multilayer Perceptron (MLP) fusion module, followed by a prediction layer for multimodal classification. We did not apply preprocessing steps and fed the input representations provided with the dataset into the transformer models. All the models were fine-tuned on the training data and evaluated on the development set provided by the shared task organizers.

We carried out our experiments on the Google Colaboratory platform with an NVIDIA L4 GPU, utilizing the PyTorch framework and AutoGluon library (Shi et al., 2021). We set uniform hyperparameter settings for all the examined models: a base learning rate of 1e-4, decay rate of 0.9 using cosine decay scheduling, batch size of 8, maximum training epochs of 10, and optimization via the AdamW optimizer. Each model was trained for around 12 minutes, regardless of the subtask.

3.1 Unimodal experiments: textual modality

Our language model selection strategy was inspired by previous studies that showed that language-specific models pre-trained with larger vocabulary and bigger language-specific datasets usually outperform multilingual models such as mBERT (Virtanen et al., 2019). Consequently, our experiments focused on comparing several state-of-the-art Arabic language models:

- **MARBERT** (Abdul-Mageed et al., 2021): a transformer-based model pre-trained on a large corpus of 1 billion Arabic tweets, which predominantly includes modern standard Arabic and various dialectal forms, thus enhancing its language understanding capability across various Arabic dialects. The model was pre-trained using the same architectural

framework as BERT (Devlin et al., 2019), focusing exclusively on the masked language modeling (MLM) objective but omitting the next sentence prediction (NSP) component. The large amount of tweets in the training data makes the model particularly suited for processing short social media texts.

- **CAMeLBERT** (Inoue et al., 2021): a collection of BERT models pre-trained on Arabic texts with different sizes and variants, including modern standard Arabic, dialectal Arabic, and classical Arabic. Recently, the authors proposed a new version called CAMeLBERT-MIX-SA, which is pre-trained on a mix of 167GB of texts in the aforementioned three Arabic variants and fine-tuned on the ASTD (Nabil et al., 2015), ArSAS (AbdelRahim Elmadany and Magdy, 2018), and SemEval datasets (Rosenthal et al., 2017). We used the CAMeLBERT-MIX-SA version in our experiments.
- **GigaBERT** (Lan et al., 2020): a customized bilingual BERT model designed for zero-shot transfer learning from English to Arabic. The model is pre-trained on a multilingual corpus that includes 6.1 billion tokens in English and 4.3 billion in Arabic, sourced from the Gigaword corpora (Parker et al., 2011), Wikipedia, and the OSCAR corpus (Suárez et al., 2019). The authors customized the vocabulary and augmented their data with code-switched samples to improve cross-lingual performance. These modifications enabled GigaBERT to achieve superior performance compared to other multilingual BERT-based models on various Arabic natural language processing tasks, such as named entity recognition and part-of-speech tagging.

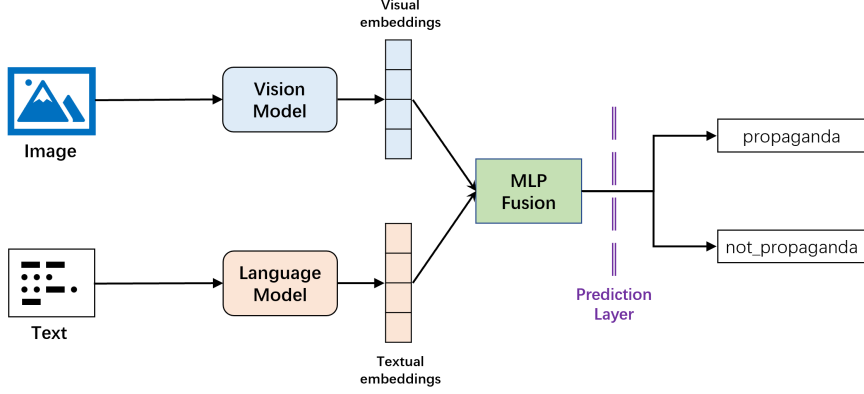


Figure 1: An overview of the multimodal classification system.

3.2 Unimodal experiments: visual modality

We examined two visual models for classifying the visual modality, i.e., meme’s image only:

- **EVA** (Fang et al., 2023): a vision-centric foundation model designed to explore the limits of visual representation at scale using only publicly accessible data. It introduces masked image modeling (MIM) pre-training task that reconstructs masked-out image-text aligned vision features based on visible image patches. This enabled EVA to efficiently scale up to one billion parameters and achieve state-of-the-art performance across various downstream visual tasks (e.g., image and video classification, object detection) without extensive supervised training.
- **CAFormer** (Yu et al., 2024): a vision model that incorporates a mix of depthwise separable convolutions and vanilla self-attention mechanisms within a MetaFormer architecture, which allows the model to effectively manage computational complexity while capturing long-range dependencies. The model achieved a top-1 accuracy of 85.5% on ImageNet-1K (Russakovsky et al., 2015), demonstrating its effectiveness under supervised training conditions without the need for external data or distillation.

3.3 Multimodal experiments

We employed a multimodal architecture that integrates language and visual models, serving as text and image encoders to extract contextualized embeddings from textual and visual inputs. The resulting embeddings are concatenated using the Multilayer Perceptron (MLP) fusion module (Shi et al., 2021), where the top vector representations

from the different models are combined into a single vector. A prediction layer is subsequently added to classify each instance into one of the predefined categories: `propaganda` or `not_propaganda`. Figure 1 illustrates the details of this multimodal architecture, which has proven effective for detecting other types of harmful multimodal content, such as multimodal hate speech (Wang and Markov, 2024b) and fine-grained types of hateful memes (Wang and Markov, 2024a).

4 Results

We report the results obtained on the development and test sets in terms of the official evaluation metric: macro-averaged F1 score.

The results for subtask 2A are provided in Table 2. The MARBERT model outperformed the other examined transformer models on the development set by 1–2 F1 points. On the test set, this model showed a drop of about 2.5 F1 points, achieving an F1 score of 77.96%, which is marginally (0.73 F1 points) lower than the best-performing system in this subtask.

Table 2: Results for subtask 2A on the dev and test sets.

Set	Language model	macro-F1
Dev	MARBERT	80.48
	CAMeLBERT	79.32
	GigaBERT	78.66
Test	MARBERT	77.96

The results for subtask 2B are shown in Table 3. It can be observed that the EVA model showed higher performance than CAFormer on the development set. We once again observe a drop in performance when the model is evaluated on the test

set (ca. 6 F1 points). Nonetheless, the results obtained on the test set for this subtask are substantially higher than those of the second-runner, with a difference of 4.7 F1 points.

Table 3: Results for subtask 2B on the dev and test sets.

Set	Vision model	macro-F1
Dev	EVA	76.89
	CAFormer	75.23
Test	EVA	71.04

The performance comparison results for subtask 2C are detailed in Table 4. Although MARBERT was the best-performing language model in isolation, GigaBERT combined with EVA showed higher performance after multimodal fusion. We submitted the multimodal model that showed the best results on the development set for further evaluation on the test set and achieved an F1 score of 79.80, which is very close to the second place (79.87 F1 score) and only 0.71 F1 points away from the first place (80.51 F1 score).

Table 4: Results for subtask 2C on the dev and test sets.

Set	Multimodal model	macro-F1
Dev	GigaBERT + EVA	82.60
	MARBERT + EVA	81.59
	CAMeLBERT + EVA	80.79
	MARBERT + CAFormer	80.53
	GigaBERT + CAFormer	80.26
	CAMeLBERT + CAFormer	77.52
Test	GigaBERT + EVA	79.80

The obtained results indicate that the models trained on texts extracted from memes show higher performance than the models trained on the visual modality, implying the importance of textual modality for detecting propagandistic memes. When both modalities are combined, there is a further increase in performance by around 2 F1 points.

We provide confusion matrices for the submitted models on the test set for all three subtasks in Figures 2, 3, and 4. The multimodal model for subtask 2C exhibited the best performance and the lowest false positive and false negative rates in propagandistic memes classification. Conversely, the vision model for subtask 2B performs the worst among the three, with the lowest number of true positives (98) and the highest number of false negatives (73).

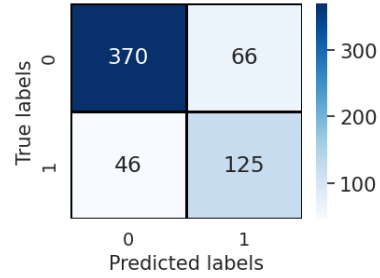


Figure 2: Confusion matrix for the MARBERT model on the test set for subtask 2A (0 = not_propaganda, 1= propaganda).

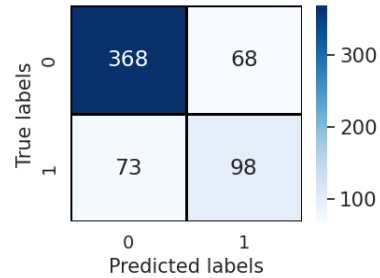


Figure 3: Confusion matrix for the EVA model on the test set for subtask 2B (0 = not_propaganda, 1= propaganda).

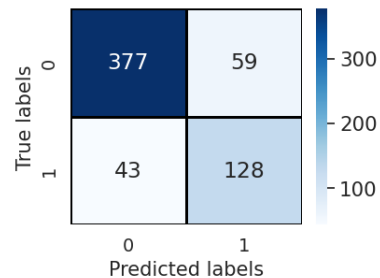


Figure 4: Confusion matrix for the multimodal model on the test set for subtask 2C (0 = not_propaganda, 1= propaganda).

5 Conclusion

In this paper, we presented the CLTL system developed for the ArAIEval 2024 Shared Task on multimodal propagandistic memes classification. Our approach involved leveraging state-of-the-art transformer models for both textual and visual modalities and employing the Multilayer Perceptron fusion module to combine text and image representations for multimodal classification. Our system secured 2nd place with a 77.96% macro-F1 score for subtask 2A, 1st place with 71.04% for subtask 2B, and 3rd place with 79.8% for subtask 2C.

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